



IILM UNIVERSITY

INTERNSHIP REPORT

Rule-Based AI Order Support Bot

Student Name	Ameer Hamza
Enrollment Number	25scs1003004679
Program & Semester	B.Tech – Computer Science & Engineering, 2nd Year, 3rd Semester
Batch	2cse10
Organization Name	DecodeLabs AI

CERTIFICATE

This is to certify that

Ameer Hamza

of B.Tech Computer Science & Engineering, IILM University, Greater Noida, has completed the internship project titled “Rule-Based AI Order Support Bot” as part of the internship/project requirements. The work presented in this report describes the design, implementation and testing of a Python-based order support chatbot developed using rule-based logic and a CSV order dataset.

DECLARATION

I, Ameer Hamza, hereby declare that this internship report titled “Rule-Based AI Order Support Bot” is prepared as part of my academic internship/project work. The report presents the work carried out during the internship and is written for academic submission. Wherever external concepts, libraries or references have been used, they have been acknowledged appropriately.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to IILM University, Greater Noida, for providing me with the opportunity to undertake this internship project. I am thankful to my faculty members and project guide for their guidance, suggestions and encouragement throughout the work. I also thank DecodeLabs AI for providing an internship environment in which I could apply programming and data handling concepts to a practical AI-based problem. Finally, I am grateful to my family and friends for their continuous support.

ABSTRACT

This internship project presents a Rule-Based AI Order Support Bot developed in Python. The main objective is to provide quick responses to common order-related customer queries without requiring a complex machine-learning model. The chatbot uses predefined intents, keyword matching and a CSV-based order dataset to answer questions such as order-status lookup, tracking-number lookup, customer order history, total spending and coupon-related information. The system runs through a continuous conversational loop and includes greetings, exit commands and a fallback response for unknown requests. Python and pandas are used for data processing, while dictionary-based logic is used to map user messages to appropriate actions. The project demonstrates how simple AI techniques can solve practical customer-support problems and provides a foundation for future improvements such as NLP classification, multilingual support, a web interface and database integration.

Keywords

Artificial Intelligence, Rule-Based Chatbot, Python, Pandas, Customer Support, Order Management, CSV Dataset

TABLE OF CONTENTS

1. Introduction
2. Organization / Internship Overview
3. Problem Statement
4. Objectives
5. Scope of the Project
6. Technologies and Tools Used
7. Dataset Description
8. System Methodology
9. System Design and Workflow
10. Implementation
11. Testing and Results
12. Advantages and Limitations
13. Future Enhancements
14. Learning Outcomes
15. Conclusion
16. References

1. INTRODUCTION

1.1 Background

Customer support systems are important for online businesses because customers frequently need information about orders, delivery status, tracking numbers, previous purchases and payments. A chatbot can reduce repetitive manual work by providing instant answers to common questions. For a beginner-level AI internship, a rule-based chatbot is a useful way to understand the basic ideas behind conversational systems, intent recognition and data-driven responses.

1.2 Project Overview

The project developed during the internship is a Rule-Based AI Order Support Bot. The bot accepts text from the user, identifies the likely intent using predefined rules and then retrieves relevant information from an orders.csv dataset. It is designed to be simple, understandable and runnable in a Python environment such as VS Code or Google Colab.

2. ORGANIZATION / INTERNSHIP OVERVIEW

2.1 Internship Context

The internship project focused on applying Python programming and introductory artificial intelligence concepts to a practical customer-support problem. The work involved understanding a project brief, preparing a dataset-driven solution, implementing chatbot logic, testing different types of inputs and documenting the final system.

2.2 Internship Project

The selected project was Project 1: Rule-Based AI Order Support Bot. The project emphasizes logical decision-making rather than advanced machine learning. This makes the system easy to test and explain while still demonstrating a practical AI-style interaction.

3. PROBLEM STATEMENT

3.1 Problem

Customer-support teams receive many repetitive questions. Manually answering every query can consume time and may lead to delays. A lightweight automated assistant is required to provide consistent responses for common order-related questions.

3.2 Proposed Solution

The proposed solution is a Python chatbot connected to an order dataset. The chatbot recognizes predefined intents and performs operations such as searching by order ID or tracking number, showing customer order history, calculating total spending and explaining coupon information.

4. OBJECTIVES

Primary Objective

To develop a simple rule-based AI chatbot that can respond to predefined customer order-support queries using Python and an order dataset.

Specific Objectives

- Read and process order information from a CSV file.
- Recognize common user intents using keyword/dictionary logic.
- Search order records by order ID and tracking number.
- Display customer order history and calculate total spend.
- Provide coupon-related explanations.
- Handle greetings, exit commands and unknown queries.
- Create a beginner-friendly and explainable AI application.

5. SCOPE OF THE PROJECT

5.1 Included Scope

The system covers common order-support tasks using structured data. It can be extended to additional intents, more order fields and a graphical or web interface.

5.2 Out of Scope

The current prototype does not perform real-time courier API tracking, natural-language model training, payment processing, voice interaction or automatic customer messaging. These can be considered future enhancements.

6. TECHNOLOGIES AND TOOLS USED

Python

Python is used as the main programming language because it is simple, readable and provides strong support for data processing and AI-related development.

Pandas

Pandas is used to read the CSV dataset, filter records, group customer orders and calculate values such as total spending.

CSV Dataset

The order data is stored in CSV format. This makes the prototype lightweight and easy to test.

VS Code / Google Colab

The project can be developed and executed in a local Python environment such as VS Code or in Google Colab.

7. DATASET DESCRIPTION

7.1 Dataset Fields

The sample order dataset contains fields such as Order ID, Date, Status, Fulfilment, Sales Channel, ship-service-level, Style, SKU, Category, Size, ASIN, Courier Status, Qty, currency, Amount, ship-city, ship-state, ship-postal-code, ship-country, promotion-ids, B2B and fulfilled-by.

7.2 Data Usage

The chatbot uses these fields to answer user questions. For example, Order ID can be used for direct lookup, Amount can support spending calculations, Status can describe order progress and customer/order identifiers can be used to retrieve purchase history.

8. SYSTEM METHODOLOGY

8.1 Input

The user enters a natural-language-style text query such as “Where is order 123?” or “Show my order history.”

8.2 Intent Detection

The system checks the message against predefined keywords and patterns. Examples include words related to order status, tracking, history, spending, coupons, greetings and exit commands.

8.3 Data Retrieval

After identifying an intent, the relevant function searches the pandas DataFrame and prepares the requested result.

8.4 Response Generation

The chatbot converts the retrieved information into a short response and displays it to the user.

8.5 Fallback

If the input does not match any supported intent, the bot returns a fallback response asking the user to try a supported query.

9. SYSTEM DESIGN AND WORKFLOW

9.1 Workflow

Start → Load orders.csv → Display welcome message → Take user input → Check exit/greeting → Detect intent → Run relevant function → Search/process dataset → Generate response → Repeat until exit.

9.2 Major Functions

Typical functions include order ID lookup, tracking-number lookup, customer order history, total-spend calculation and coupon explanation. A continuous loop manages the conversation.

10. IMPLEMENTATION

10.1 Dataset Loading

The orders.csv file is loaded into a pandas DataFrame. Basic checks are performed so that column names and missing values can be handled safely.

10.2 Rule-Based Logic

A dictionary or if-else based intent structure maps keywords to actions. For example, the presence of terms such as “track” or “tracking” can route the query toward the tracking lookup function.

10.3 Order Search

Order ID searches filter the DataFrame for a matching identifier. The returned record can include status, courier information, amount and other available fields.

10.4 Customer History

Customer-related queries can be processed by filtering records associated with the customer and presenting previous orders in a readable format.

10.5 Total Spend

The amount column can be converted to numeric form and summed for the selected customer's records. This demonstrates basic aggregation using pandas.

10.6 Conversation Loop

The chatbot continues accepting queries until the user enters an exit command such as “exit”, “quit” or “bye”.

11. TESTING AND RESULTS

11.1 Test Cases

1. Greeting input → Bot responds with a welcome message.
2. Valid order ID → Bot returns the matching order details.
3. Valid tracking number → Bot returns available tracking information.
4. Customer history request → Bot lists matching previous orders.
5. Total-spend request → Bot calculates the total amount.
6. Coupon query → Bot explains available coupon information stored/defined by the system.
7. Unknown input → Bot gives a fallback response.
8. Exit command → Bot ends the session.

11.2 Expected Result

The prototype successfully demonstrates the complete flow from user input to rule selection, dataset search and response generation. Because the system is rule-based, supported queries can be handled consistently and the logic remains easy to understand and debug.

12. ADVANTAGES AND LIMITATIONS

Advantages

- Simple and beginner-friendly.
- Fast response for supported queries.
- Easy to explain and debug.
- Does not require model training.
- Works with a lightweight CSV dataset.
- Can be extended with new rules and functions.

Limitations

- Understanding is limited to predefined rules.
- Different wording may not always be recognized.
- No real-time courier integration.
- CSV is less suitable than a database for large-scale deployment.
- No advanced NLP, voice or multilingual capability in the basic prototype.

13. FUTURE ENHANCEMENTS

Planned Improvements

The system can be upgraded with NLP-based intent classification, fuzzy matching, multilingual support, a web/mobile interface, a database such as MySQL, real-time courier APIs, authentication, conversation history and analytics dashboards. A machine-learning or transformer-based model could also improve understanding of varied user language.

14. LEARNING OUTCOMES

Technical Learning

During the project, I strengthened my understanding of Python functions, loops, conditional logic, dictionaries, file handling and pandas DataFrames. I also learned how structured data can be connected to an interactive application.

AI Learning

The project introduced practical concepts of intent detection, rule-based reasoning and automated response generation. It showed how AI-style behavior can be implemented even before moving to advanced machine-learning models.

Professional Learning

The internship improved problem-solving, debugging, documentation and project-presentation skills. It also provided experience in converting a project brief into a working prototype.

15. CONCLUSION

Conclusion

The Rule-Based AI Order Support Bot is a practical and lightweight prototype for automating repetitive customer order queries. It combines Python programming, pandas-based data processing and rule-based intent detection to create an interactive support assistant. The project successfully demonstrates the basic pipeline of input processing, intent recognition, data retrieval and response generation. Although the current system has limitations, its modular design makes it suitable for future development into a more advanced customer-support platform.

16. REFERENCES

References

1. Python Documentation – Python programming language reference and standard library.
2. Pandas Documentation – Data analysis and DataFrame operations.
3. Google Colab Documentation – Cloud-based Python notebook environment.
4. General concepts and project brief supplied as part of the internship assignment.
5. Dataset documentation/metadata associated with the provided orders.csv file.

APPENDIX A – PROJECT SNAPSHOT

A typical execution starts by loading the orders.csv file, validating the required columns, and then entering the chatbot loop. The user can ask supported order questions and receive responses generated from the dataset.

APPENDIX B – SAMPLE INTERACTIONS

Speaker	Sample Interaction
User	Hello
Bot	Hello! How can I help you with your order today?
User	Track my order
Bot	Please provide your order ID or tracking number.
User	Show my order history
Bot	The bot searches the dataset and displays matching order records.
User	How much have I spent?
Bot	The bot calculates the total amount from matching records.
User	Bye
Bot	Thank you. Have a great day!